Action Items: MaskClass — Pre-Submission Checklist for NeurIPS 2026

CRITICAL: Paper Will Be Rejected Without These

1. Convert to NeurIPS 2026 LaTeX Template

What: The paper currently uses the UAI 2026 two-column template. NeurIPS requires single-column with `neurips_2026.sty`.
How: Download the NeurIPS 2026 template from the official website. Migrate all content. Ensure the paper fits within the 9-page content limit (references and appendix are unlimited). The footer "Submitted to the 42nd Conference on Uncertainty in Artificial Intelligence" must be removed.
Effort: ~4 hours.

2. Add Real-Data Experiments (3 Datasets Minimum)

What: All three simulated reviewers unanimously flag synthetic-only evaluation as disqualifying.
How:

- **PBMC 68k** (Zheng et al., 2017): Download from 10x Genomics. ~68,000 cells, 11 annotated types.
- **Baron human pancreas** (GEO: GSE84133): ~8,500 cells, 14 cell types.
- **Zeisel mouse cortex** (GEO: GSE60361): ~3,000 cells, 7 major types.
- For each: run MaskClass + all baselines (DCA, MAGIC, AutoClass, ccImpute, SAVER, scVI, ALRA).
- **Metrics (no ground truth):** Leiden clustering → ARI and NMI vs. published labels; gene-gene correlation recovery (protocol from MAGIC paper); differential expression concordance vs. bulk reference (if available).
- Add results as new Section 4.2 + Table 3.
  Effort: 2–3 days including data download, preprocessing, and running all methods.

3. Add Modern Baselines: scVI and ALRA

What: The current baseline set (DCA 2019, MAGIC 2018, SAVER 2018, etc.) is outdated. scVI is the most widely used deep learning method; ALRA is a zero-preserving imputation method by authors already cited.
How:

- **scVI:** `pip install scvi-tools`. Use `scvi.model.SCVI` with default parameters. Train on each dataset. Extract denoised expression via `model.get_normalized_expression()`.
- **ALRA:** Use the R package (`devtools::install_github("KlugerLab/ALRA")`) or Python port. Run with default rank selection.
- Add both to all tables (synthetic + real data).
  Effort: 1–2 days.

4. Add Ablation Study of All Three Proposed Components

What: The paper proposes three components (differential masking, biozero regularization, post-hoc shrinkage) but never ablates them individually.
How: Create and run 4 variants:

1. **No shrinkage:** Set $\alpha = 0$ in the shrinkage step.
2. **No biozero regularization:** Set $\lambda_{bio} = 0$.
3. **Uniform masking:** Set $p_{zero} = p_{nz}$ (e.g., both = 0.03).
4. **Plain masked autoencoder:** All three removed simultaneously. Run each on the 13 synthetic test-split scenarios (5 seeds). Present as a new table (Table 4 or supplementary). Discuss which component drives which improvement.
   Effort: 1–2 days of GPU time.

5. Fix De-Anonymization Violation

What: Section 2.2 says "Our previous work, ccImpute [Malec et al., 2022]" — this reveals author identity in a double-blind review.
How: Change to "ccImpute [Malec et al., 2022] offers an alternative that builds a consensus similarity matrix..." Remove all instances of "our previous work," "our prior work," or similar phrases.
Effort: 15 minutes.

HIGH PRIORITY: Significantly Improves Acceptance Chances

6. Fix the p_bio Mathematical Error (Sec. 3.5)

What: The numerator of the biozero probability formula is $(1-\delta_{ij})f_{ij}(0)$, but it should be $f_{ij}(0)$ because when $t_{ij} = 0$, the observation is zero regardless of dropout.
How: Fix the formula in the code and in the paper. Re-run all experiments with the corrected formula. Report whether results change (likely minimal impact, but must be verified).
Effort: ~0.5 days.

7. Add Statistical Significance to Tables

What: Table 1 reports means but no standard deviations. Gains may not be statistically significant.
How: You already have 5-seed data. Add ± std columns to Tables 1 and 2. Optionally, run a paired Wilcoxon signed-rank test (MaskClass vs. each baseline across 13 scenarios) and report p-values.
Effort: ~2 hours to extract from existing data.

8. Add Per-Scenario Results Table

What: "Wins on 12/13 scenarios" is claimed but unverifiable — no per-scenario table exists.
How: Create a supplementary table: rows = 13 scenarios, columns = methods, entries = MSE ± std. Identify the 1 scenario where MaskClass loses and explain why.
Effort: ~2 hours (data already exists).

9. Add Downstream Task Evaluation

What: Denoising metrics alone do not prove practical value.
How:

- **Clustering (synthetic):** Run Leiden on imputed matrices, compute ARI/NMI vs. known group labels.
- **Trajectory (synthetic):** Run DPT on trajectory scenarios, compute Kendall's τ with ground-truth pseudotime.
- **Clustering (real):** Already covered in item #2. **Effort:** ~1 day (partially overlaps with item #2).

10. Add Hyperparameter Sensitivity Analysis

What: 7+ hyperparameters are fixed without justification.
How: On one representative scenario (e.g., groups_balanced_moderate_drop), vary:

- $\alpha \in \{0, 0.1, 0.2, 0.3, 0.5, 0.7, 1.0\}$
- λ_bio $\in \{0, 0.5, 1.0, 2.0, 5.0\}$
- p_zero $\in \{0.005, 0.01, 0.02, 0.05\}$ Plot MSE vs. each parameter. Present as a supplementary figure.
  Effort: ~1 day of GPU time.

11. Remove or Qualify "Scalable" from Title

What: Figure 3 shows MaskClass is the second-slowest method. The title claim is indefensible.
How: Change title to "MaskClass: A Masked Denoising Autoencoder for Accurate Single-Cell

RNA-seq Analysis." If you can implement mini-batch inference or FP16 training to achieve a real speedup, you could keep "Scalable," but this requires evidence.
Effort: 5 minutes for the title change; days if optimizing runtime.

NICE TO HAVE: Strengthens Paper Further

12. Add a Hero Figure on Page 1

What: A skimming reviewer needs to understand the contribution from title + abstract + Figure 1 alone. Currently, Figure 1 is on page 6.
How: Create a compact version of Figure 1 (or a new teaser figure) showing: (left) the zero ambiguity problem, (center) MaskClass's approach, (right) a key result comparison. Place on page 1 or 2.
Effort: ~3 hours.

13. Tighten the Abstract

What: The abstract is ~200+ words and buries contributions. NeurIPS abstracts should be ~150 words.
How: Remove cross-modality claims (unsubstantiated). Lead with the problem, state the method in 1–2 sentences, give a key quantitative result. Remove "significantly" (no statistical test).
Effort: ~30 minutes.

14. Fix Notation Conflicts

What: d = bottleneck dimension AND d_ij = dropout indicator. B = mini-batch AND biozero set. w = shrinkage weight AND loss weight.
How: Use d_lat for bottleneck, keep d_ij for dropout. Use B for mini-batch and Z_bio for biozero set. Use w^(s) for shrinkage weights.
Effort: ~1 hour (search-and-replace).

15. Move Simulation Details to Supplementary

What: Section 3.1 is very dense (~1.5 pages of Splatter configuration). This crowds out space for real-data results.
How: Keep a 0.5-page summary of the simulation setup in the main paper. Move detailed parameters, scenario catalog, and normalization details to Supplementary.
Effort: ~1 hour.

16. Fix Minor Typographical Issues

- Remove dangling "m" in Sec. 3.4 masking equation.
- Fix "Algorithm 1 (Algorithm 1)" → "Algorithm 1" in Sec. 3.9.

Fix Algorithm 1 (Algorithm 1) → Algorithm 1 in Sec. 3.5.
- Align Figure 1 caption p^(0)_bio notation with main text (remove the superscript (0) or add it to main text).
- Fix Figure 2 biozero-MSE panel y-axis (remove negative values).
- Update Figure 1 training objective to include L_nz.
- Define "Baseline" explicitly in Table 1 caption.
- State GPU hardware used for runtime comparison.
  Effort: ~1 hour total.

17. Add NeurIPS Paper Checklist

What: NeurIPS requires a reproducibility/ethics checklist (typically a separate section at the end).
How: Download the NeurIPS 2026 checklist template. Fill in all items. Many are already satisfiable (open-source code, fixed seeds, etc.).
Effort: ~1 hour.

18. Discuss gNRMSE > 1 Explicitly

What: MaskClass achieves gNRMSE = 1.17, meaning reconstruction error exceeds gene-level std on average. This is not discussed.
How: Add 2–3 sentences in Sec. 4.1 contextualizing this: (a) the metric is strict because it includes low-variance genes where even small errors appear large relative to the gene's variability; (b) MaskClass still substantially improves over all baselines on this metric.
Effort: ~15 minutes.

VERIFY: Items Flagged During Anti-Hallucination Sweep

19. Verify "wins on 12/13 scenarios" claim

Action: Extract per-scenario MSE from your experimental logs. Count how many scenarios have MaskClass as the lowest-MSE method. If the number is not 12, correct the claim.

20. Verify "state-of-the-art" characterization of baselines

Action: After adding scVI and ALRA, you can more defensibly call the comparison "representative." However, consider also searching for 2024–2025 scRNA-seq denoising methods (check NeurIPS 2024, ICML 2024, Nature Methods 2024–2025) to ensure nothing major is missing.

21. Verify cross-modality applicability claim

Action: Either (a) remove the claim from the abstract and paper, or (b) run MaskClass on at least

Action: Either (a) remove the claim from the abstract and paper, or (b) run MaskClass on at least one non-scRNA-seq dataset (e.g., CITE-seq protein counts, spatial transcriptomics) and report results. Option (a) is safer and faster.

22. Verify runtime hardware consistency

Action: Confirm all methods in Figure 3 were benchmarked on the same GPU hardware. State the GPU model (e.g., "NVIDIA A100 40GB") in the paper.

Summary Timeline Estimate

Priority	Items	Estimated Total Effort
Critical (#1–5)	Template, real data, baselines, ablation, de-anonymize	~5–7 days
High (#6–11)	Math fix, stats, per-scenario, downstream, sensitivity, title	~3–4 days
Nice to Have (#12–18)	Hero figure, abstract, notation, supplementary, minor fixes, checklist, gNRMSE	~1–2 days
Verify (#19–22)	Claim verification	~0.5 days
Total		**~10–14 days of focused work**

End of Action Items.